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Measurement of polyethylene pellets near the glass transition temperature to enhance Raman spectral selectivity among samples and improve accuracy for density determination

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A simple and effective strategy for improving the accuracy of the multivariate determination of polyethylene (PE) density using Raman spectroscopy has been demonstrated. This strategy is based on the possibility that varied polymeric structures of the PE samples, especially at a sub-zero temperature range, would enhance their spectral selectivity, thereby potentially improving the multivariate correlation with their pre-determined physical properties such as density. For the evaluation, Raman spectra were collected at regular intervals during continuous increase of the PE temperature from cryogenic to near room temperature. Then, using partial least squares (PLS) regression, calibration models were developed to correlate the Raman spectral features collected at each time period with the reference PE density values. Interestingly, the accuracy was improved when the temperature of the PE pellets was −35 °C, near the glass transition temperature (T_g). To explain the improved accuracy, a two-dimensional (2D) correlation analysis was employed to detail the spectral variation induced by temperature change. Diverse segmental chain motions (so called micro-Brownian motion) predominantly occurring in the amorphous section of the PEs around T_g greatly enhanced the spectral selectivity among PE samples. In addition, minor β-relaxation occurring around this temperature was an additional source of the enhanced spectral selectivity. In parallel, differential scanning calorimetry (DSC) curves of the samples were also examined to check the existence of the phase transitions.

Introduction

Previously, we searched for an optimal temperature of 70 °C to provide an improved accuracy for Raman spectroscopic determination of the density of polyethylene (PE) pellets.[1] The origin for the improved accuracy was the enhanced spectral selectivity among the samples due to their diverse structural variations in the pre-melt stage at the found temperature, and Raman spectroscopy was eventually effective to recognize phase transitions of the polymer such as pre-melting. The demonstrated strategy was simple since the spectral selectivity could be elevated just by maintaining the sample temperature at a condition that could induce transient variation of the polymeric structure.

When the temperature of a polymer alternatively changes from the sub-zero to the cryogenic range, different types of phase transitions such as the glass transition could occur[2,3] and this could also possibly enhance structural selectivity among the samples.[4] The nature of the glass transition and the existence of dynamic divergence near the glass transition temperature (T_g) have been studied by several research groups.[5,6] For example, Xu *et al.* explored the nature of the dynamics in the glass transition with the use of a Dyre shoving model to explain the temperature dependence of polymer viscosity.[7] Wu *et al.* studied the dynamic response of the glass forming systems based on dynamic mechanical measurements.[8] The major challenge in these researches was to overcome the difficulty in obtaining reliable dynamic properties of a sample near and below T_g, due to an extremely long equilibrium relaxation time at these conditions. Owing to this fact, the essence of the glass transition has not yet been fully understood. Therefore, studies of diverse analytical methods able to provide further insight on the glass transition are beneficial.

In this publication, we have examined Raman spectral variations of PE pellets in the sub-zero to cryogenic temperature

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range to study possible phase changes such as the glass transition and to simultaneously search for conditions that could also enhance spectral selectivity among the samples, to improve the accuracy of their density determination using multivariate analysis. Spectral variations occurring in the sub-zero to cryogenic temperature range are also expected to be different from sample to sample depending on the individual polymeric structures; therefore, a certain state leading to enhanced spectral selectivity among the samples would exist analogously. For this purpose, the PE pellets were rapidly cooled to a cryogenic temperature by simply placing a glass vial containing the sample in liquid nitrogen. Then, the Raman spectra of the samples were collected at regular intervals while the sample temperature increased naturally to near ambient temperature. By this method, the Raman spectra of the PE pellets over a wide temperature range could be easily collected without the use of any extensive peripheral devices.

The changes in the PE band intensity, position and width as a function of temperature were initially examined in detail. Then, partial least squares (PLS) regression[9,10] was used to determine the density of the PE pellets using the spectra collected at each time period, and then the resulting accuracies were compared to identify a temperature range providing improved accuracy. It is necessary to clarify, to avoid confusion, that this study intends not to measure the actual densities of PEs at sub-zero temperatures but to probe for a Raman measurement temperature capable of enhancing spectral discriminability among the samples, to improve multivariate correlation with their density values which were predetermined using a standard ASTM method (density measurement at 23 °C).

Two-dimensional (2D) correlation analysis[11–13] was used to assess the Raman spectral variation of the samples according to the changes in reference density as well as in the temperature. 2D correlation analysis is an effective method for investigating structural variation when external perturbations are applied to samples, as is generally known. Finally, the results of the 2D correlation analysis were examined to justify the improved accuracy of Raman measurements performed within the selected temperature range.

Experimental section

Sample details, differential scanning calorimetry (DSC) analysis and collection of Raman spectra

The 25 PE pellets adopted in a previous publication were also used in this study.[1,14] The samples were composed of three different grades (homo PE, 1-butene copolymerized PE, and 1-octene copolymerized PE) with a density range of 0.918–0.961 g cm^{-3}. The reference density of each sample was determined using a standard method described in ASTM D1505-3.[15] The PE pellets were simply transferred into a circular glass vial (diameter: 9.8 mm) for collection of Raman spectra, as shown in Fig. 1.

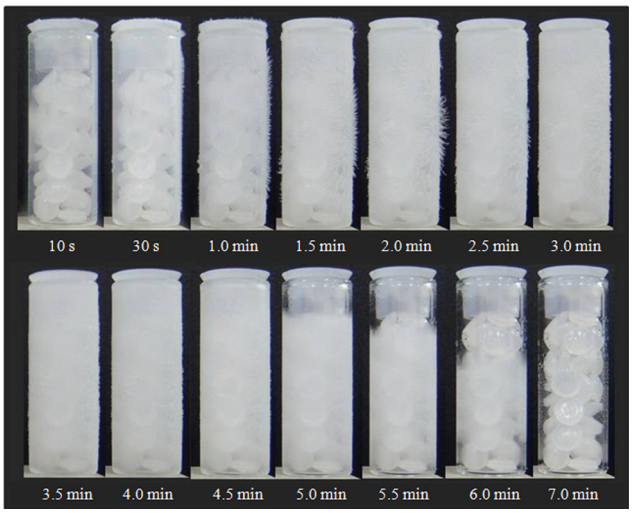

Fig. 1 Pictures of the low density PE pellets (density: 0.918 g cm^{-3}) in a glass vial, during temperature elevation from cryogenic to near room temperature.

DSC curves of the samples were obtained over a temperature range of −179 to 15 °C using a TA Instrument model DSC2910 system (New Castle, DE, USA). Approximately 6.5 mg of the sample taken from each pellet was transferred into a standard DSC aluminum pan for analysis. The initial temperature was set to −179 °C, to cool the sample over 15 minutes, and the sample was then heated at a rate of 30 °C per minute until the final temperature reached 15 °C. An empty pan was used for the reference.

Raman spectra were collected using the wide illumination (WAI) scheme (PhAT system, Kaiser Optical Inc., Ann Arbor, MI, USA), as previously described.[1,14] It was capable of illuminating a laser (785 nm) onto a large sample area with a diameter of 6 mm (area: 28.3 mm^2). The vial containing the PE pellets was initially immersed in liquid nitrogen to drive the sample temperature to a cryogenic level. After temperature equilibration for 15 minutes, the vial was taken out quickly and immediately positioned at the focal point of the WAI scheme for acquisition of spectra. While the temperature of the sample was naturally increasing to near ambient temperature, Raman spectra were continuously collected at intervals of 20 seconds (laser exposure time: 20 seconds) and a total of 25 spectra (or snapshots) were collected for each sample.

The temperature of the sample was checked by inserting a thermocouple into the pellet packing. Triplicate spectra were collected for each sample. The same PE pellets were divided into three different vials, and then corresponding Raman spectra were separately collected as described above. The resolution of the collected spectra was 4 cm^{-1}. All spectral preprocessing and multivariate methods, including baseline correction, normalization, PLS and 2D correlation analysis, were performed using Matlab version 7.0 (Math Works Inc., MA, USA).

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Results and discussion

Raman spectral variations of the PE pellets at different temperatures

Fig. 1 shows pictures of the low density PE pellets (density: 0.918 g cm^{-3}) in a glass vial during temperature elevation from cryogenic to near room temperature. The time at which the picture was taken is designated on each picture. As soon as the sample vial was taken out of the liquid nitrogen, frost immediately started to form on the wall and then became thicker. The relative humidity in the lab was maintained at approximately 21.0%. The frost was thickest at 2 minutes (corresponding to the 6th spectral collection (hereinafter, snapshot), temperature around −45 °C) and became thinner thereafter. At 5 minutes (the 15th snapshot, temperature around 1 °C), it started to disappear. Fortunately, frost (basically water) is a very weak Raman-scatterer, and so its influence on the overall spectral features of the PE pellets would not be substantial. In addition, an increase of the pellet volume of approximately 9% was observed during the temperature elevation due to thermal expansion.

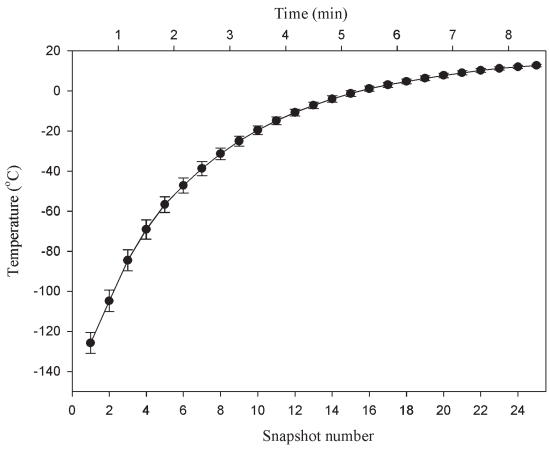

Fig. 2 Temperature variation of the PE pellets as a function of time, in which the x-axis is designated by both time and snapshot number. The curve corresponds to the average of five independent temperature measurements from five separate batches prepared from the same PE pellets. The bars in each snapshot indicate the corresponding standard deviation of the temperature measurements.

The temperature variation of the PE pellets is shown in Fig. 2, where the x-axis is designated by both time and snapshot number. The curve in the figure corresponds to the average of five independent temperature measurements from five separate batches prepared from the same PE pellets. The bars on each snapshot indicate the corresponding standard deviation of the temperature measurements. The rate of temperature elevation is faster in the earlier stages and substantially slower in the later stages. Therefore, the magnitude of the error bar is relatively larger in the earlier stages.

Fig. 3(a) shows the Raman spectra (1504–1054 cm^{-1} range) of the low density PE sample (density: 0.918 g cm^{-3}) collected from different snapshots (temperatures). Although a total of 25 Raman spectra were collected during the elevation of the temperature, only 5 Raman spectra were selected, for easier comparison of the spectral features. In addition, a Raman spectrum obtained at room temperature (23 °C) is also shown. PE Raman bands are clearly observable in each snapshot, while the baselines of the spectra vary minutely due to the presence of frost on the wall of the vial.

To compare the spectral features in each snapshot, the spectra were normalized. The baseline-corrected spectrum at

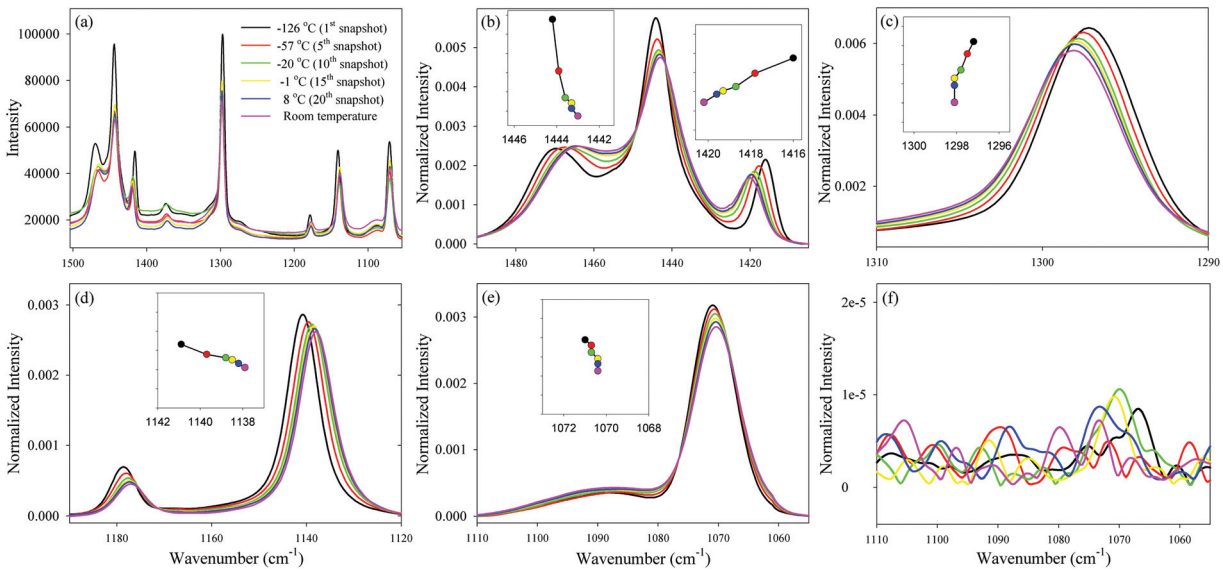

Fig. 3 (a) Raman spectra of a PE sample (density: 0.918 g cm^{-3}), collected from different snapshots in the 1504–1054 cm^{-1} range. To examine the spectral features in detail, the corresponding normalized spectra are highlighted in four different narrower spectral ranges: (b) 1490–1405, (c) 1310–1290, (d) 1190–1120 and (e) 1110–1054 cm^{-1}. Standard deviations (SDs) of the intensities in the 1110–1054 cm^{-1} range, calculated using independently collected triplicate spectra, are also shown (f).

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five different wavenumbers (1504, 1402, 1196, 1114 and 1054 cm^{-1}) was divided by the corresponding peak area under the 1504–1054 cm^{-1} range for the normalization. Four narrower spectral ranges of 1490–1405, 1310–1290, 1190–1120 and 1100–1054 cm^{-1} are highlighted in Fig. 3(b), (c), (d) and (e), respectively. For easier tracking of the spectral variation, the changes in the maximum peak positions and the corresponding intensities of the major bands are also shown in the insets. The magnitudes of the *x*- and *y*-span of the inset plots are the same. In the 1490–1405 cm^{-1} range (Fig. 3(b)), where the 1464 (CH_2 rocking), 1443 (CH_2 bending) and 1420 cm^{-1} (CH_2 bending) bands at room temperature are present,[1,14,16–21] the shapes of the bands clearly vary as the temperature increases. The 1464 cm^{-1} band shifts to 1469 cm^{-1} at −126 °C (1st snapshot). Then, it moves to a lower wavenumber, and its bandwidth broadens in the following snapshots. Conversely, the 1420 cm^{-1} band shifts to 1416 cm^{-1} at −126 °C. Then, it moves to higher wavenumbers and its bandwidth marginally broadens with temperature elevation. The 1443 cm^{-1} band shifts to a slightly higher wavenumber at −126 °C, then the corresponding intensity decreases and the bandwidth becomes broader as the sample temperature increases.

The 1298 cm^{-1} (CH_2 twisting) band at room temperature shifts slightly to 1297 cm^{-1} at −126 °C (1st snapshot), as shown in Fig. 3(c). Then, it moves to higher wavenumbers with increasing temperature. In Fig. 3(d), the 1177 (CH_2 rocking) and 1138 cm^{-1} (C–C stretching) bands show similar behavior upon temperature perturbation, as does the 1464 cm^{-1} band. In Fig. 3(e), the 1070 cm^{-1} band (C–C anti-symmetric stretching) corresponding to the crystalline component in PE and the broad spectral features around 1085 and 1065 cm^{-1} originating from the amorphous structure[16–21] are shown. With an increase in the sample temperature, the intensity of the crystalline peak decreases and the intensity of the amorphous peak synchronously increases. This indicates that the structure becomes more crystalline at very low temperatures.

A superior repeatability of the measurements is necessary to recognize minute spectral variations induced by temperature change. To assess the repeatability, standard deviations (SDs) of the intensities at each wavenumber were calculated using independently collected triplicate spectra in each case. Fig. 3(f) presents the SDs of the intensities calculated from the spectra shown in Fig. 3(e). It is important to note that the *y*-span is 100 times smaller compared to that in Fig. 3(e). The magnitudes of the SDs are very small and the average SD from these observations is 3.3×10^{-6} normalized absorbance unit (AU). In the meanwhile, the magnitude of the intensity change of the 1070 cm^{-1} peak between the measurements at −126 (1st snapshot) and 8 °C (20th snapshot) is 0.000263 normalized AU, and so the average intensity change between the adjacent snapshots is 1.38×10^{-5}. The magnitude of the average SD relative to that of the average intensity change per snapshot is 24%, so the measurement is sufficiently precise to perceive the temperature-induced spectral variations between the snapshots.

All of the spectral variations described above were more significant at lower temperature (earlier snapshots) and most of the PE bands broadened with temperature elevation. The same spectral variations shown in Fig. 3 were also examined for the middle (0.934 g cm^{-3}) and high (0.961 g cm^{-3}) density samples, although the corresponding figures are not shown here. The trends in the spectral variations for these samples were similar to that observed for the low density PE in Fig. 3, while the magnitude of spectral variation and the change in band shape upon temperature perturbation were slightly different.

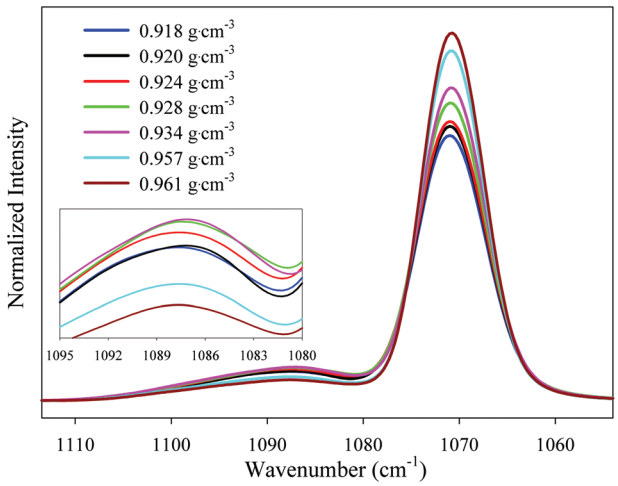

Fig. 4 Raman spectra of seven selected PE samples of low to high densities measured at −126 °C (the 1st snapshot). The inset highlights the spectral variations in the 1095–1080 cm^{-1} range.

Fig. 4 shows the Raman spectra of seven selected PE samples of low to high densities measured at −126 °C (the 1st snapshot). The inset highlights the spectral variations in the 1095–1080 cm^{-1} range. The intensity of the 1070 cm^{-1} band was higher for the high density PEs than for the low density PEs. This obviously indicates a larger fraction of crystalline component in the high density PEs. In contrast, the intensities of the amorphous bands were relatively higher for the low density PEs. Examination of the spectra in Fig. 3(e) and 4 confirms that the intensities of the crystalline and amorphous bands of the PE pellets vary with both temperature and density. Therefore, the temperature-induced spectral variations of the PE samples are expected to be highly diverse from each other. Thus, the accuracy of measurement at each snapshot could be dissimilar due to the altered spectral features of the samples caused by temperature change.

Determination of the density of the PE pellets in each snapshot

PLS[9,10] was used to determine the density of the PE pellets using the Raman spectra collected from each snapshot. Initially, 20 samples and 5 samples were assigned to the calibration and validation sets, respectively. All combinations of the selected five samples as the validation set were tested, and the mean standard error of calibration (MSEC) and the mean standard error of prediction (MSEP) were calculated by averaging the individual SEC and SEP values for each combination,

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as performed in a previous study.[8,14] The 1504–1054 cm^{-1} range shown in Fig. 3(a) was used for PLS.

The MSEPs obtained for each snapshot are displayed in Fig. 5 (as circles). The corresponding measurement temperatures are also indicated on the upper *x*-axis. In all cases, six PLS factors were used. The dashed line indicates the MSEP acquired from measurements at room temperature (designated as $MSEP_{RT}$), which was 0.00084 g cm^{-3} and is comparable to the MSEP (0.00080 g cm^{-3}) achieved in a previous study.[14] From the 1st to 4th snapshots (sample temperatures from −126 to −69 °C), the MSEP dropped sharply below the $MSEP_{RT}$ and then reached a minimum at the 7th snapshot (−39 °C), with an MSEP of 0.00068 g cm^{-3}. After the 7th snapshot, the MSEP gradually increased again. In comparison with the $MSEP_{RT}$, the improved accuracy at the 7th snapshot was statistically significant based on a *t*-test at a 95% confidence level.

Since the increase in temperature was much faster and the uncertainty in the sample temperature was larger in the lower temperature range, a doubled laser exposure time of 40 seconds for spectral acquisition could somewhat lessen the temperature uncertainty. Therefore, the spectra acquired in two adjacent snapshots were continuously averaged, and these averaged spectra were used for PLS. For example, the 1st and 2nd snapshots were averaged to yield a new 1st snapshot, and the 2nd and 3rd snapshots were averaged to generate a new 2nd snapshot. Each averaged spectrum was equivalent to spectral collection over 40 seconds of laser exposure. The MSEPs obtained for each averaged snapshot are also shown in Fig. 5 (as triangles). Six factors were also used for each case. The variation in the MSEPs obtained using two-snapshot averaged spectra was similar to that using single-snapshot spectra, while the MSEPs were slightly lower. The lowest MSEP (0.00066 g cm^{-3}) was achieved at the 7th snapshot, corresponding to the average of the original 7th and 8th snapshots. When the snapshots were further averaged, *i.e.*, by averaging three adjacent snapshots, there was no additional improvement in the MSEP. The overall results indicate that the PLS accuracy for the determination of PE density is improved by spectral acquisition around −35 °C. This finding implies that the spectral features of the PE pellets around −35 °C are more descriptive for following the variation in densities determined using the ASTM method.[22,23]

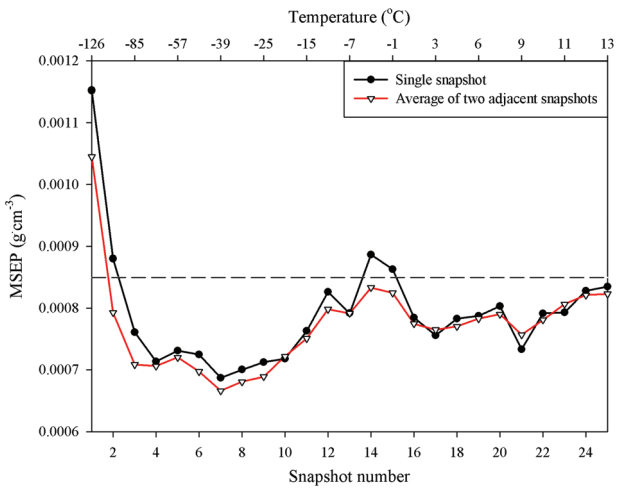

Fig. 5 The MSEP obtained for each snapshot. The circles and triangles correspond to the results obtained from the use of a single snapshot and the average of two adjacent snapshots, respectively.

2D correlation analysis

Justification for the improved accuracy around −35 °C needs to be provided. Since the temperature-induced spectral variations of the PE samples are rather complicated, two dimensional (2D) correlation analysis could be effective for examining the complexly different structural variations of the PEs induced by temperature. A detailed description of 2D correlation analysis can be found in other publications.[11–13] The 1114–1054 cm^{-1} range, containing both crystalline and amorphous bands, was used for the analysis.

Fig. 6(a) shows the synchronous map generated using all of the sample spectra of varying densities collected at −126 °C (1st snapshot). The auto-correlation peak around 1073 cm^{-1}, indicating the gradual increase in the crystalline component with the increase of sample density, is clearly shown. The development of negative cross-correlation peaks (shaded) between 1070 and 1060 cm^{-1} implies that the change in the intensity of these two bands occurred in opposite directions, confirming the simultaneous increase and decrease of the crystalline and amorphous components, respectively.

Fig. 6(b) shows the apparent asynchronous correlation peak between 1073 and 1060 cm^{-1}. This informs us that structural variation in the amorphous component occurred prior to that of the crystalline component. Such a sequential structural change suggests that the intensity variations in the crystalline and amorphous components are not complementary, in other words, no see-saw type variation occurs between these two components. Thus, the structural variations of the samples at −126 °C are nonlinear. This nonlinear behavior is probably due to the occurrence of a new component possessing a partially ordered structure. Therefore, a decrease in the amorphous structure is not linearly compensated for by an increase in the crystalline structure due to the presence of this intermediate structure. This result is consistent with previous reports by Noda *et al.*, that PE polymers undergoing transient

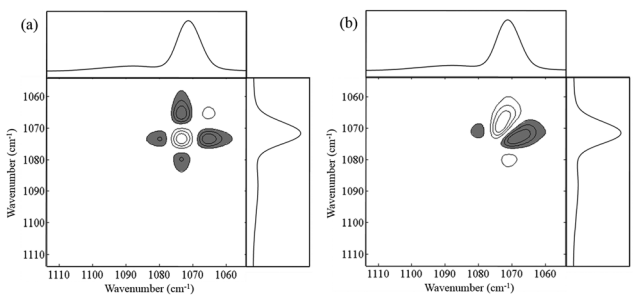

Fig. 6 Synchronous (a) and asynchronous (b) correlation maps calculated using all of the sample spectra collected at the 1st snapshot in the 1114–1054 cm^{-1} range.

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melting showed a specific phase transition from crystalline lamellae to amorphous *via* the development of a partially deformed structure.[13,24] The partially deformed structure could provide additional valuable information, which could possibly enhance spectral discrimination among PE samples and improve multivariate correlation within a target sample property.

The asynchronous correlation informs of the occurrence of new partially ordered structures; its intensity becomes lower when more of these structures are present. So, examination of the asynchronous correlation intensity is beneficial for probing the temperature-induced spectral variations, as shown in Fig. 7. The asynchronous correlation intensities in the 1073–1060 cm^{-1} range were calculated and the numbers above each circle in Fig. 7 indicate the corresponding snapshot number. The intensity gradually decreased as the sample temperature increased and the variation was rather minor after −39 °C. This implies that the diverse spectral variations occurring in the amorphous and partially ordered regions become more significant around this temperature. At an ultra-low temperature, the amorphous structure in PE is in a glassy state, where molecules are only able to vibrate marginally without distinct segmental motion. When the temperature of the sample increases further and reaches its glass transition temperature (T_g), the molecules start to wiggle and these diverse wiggling motions (so-called micro-Brownian motion) transform the sample into a rubbery state. As is generally known, the second T_g of PE is approximately around −40 °C,[25–27] which is close to the temperature providing the improved accuracy. The degree of micro-Brownian motion could differ depending on the amount of amorphous component in a sample, and also its behavior could vary from sample to sample. Thus, this is a major source for enhancing the spectral discriminability among PE samples. In addition, previous publications have shown the phase change of PEs associated with β-relaxation occurring around −35 °C,[28–30] while the resulting structural change was very small. This relaxation originates from initiation of the molecular motion of short chain branches in the interfacial regions (between the crystalline and amorphous components) of the lamellae. Therefore, it also partially contributes to the enhancement of spectral selectivity.

To alternatively confirm the structural variation around −40 °C, DSC curves of all 25 PE samples were measured and examined. Fig. 8 shows the DSC curve of the low density PE sample (0.920 g cm^{-3}) in the temperature range of −72 to −15 °C. A decrease of heat flow in the curve around −40 °C is clearly apparent and so this suggests the onset of the glass transition. Meanwhile, no meaningful changes were observed in the DSC curves of the other samples. The absence of transition peaks for these samples is acceptable because the behaviors of the coexisting amorphous and crystalline structures in a sample at glass transition are in thermally opposite directions. Glass transition can be seen as a relaxation of branches in a polymer chain that occurs mainly in the disordered amorphous structure, while the progress of relaxation is also hindered by the development of polymer chain folding resulting in an ordered lamellae structure. Since the relaxation of the branches and the chain folding are opposite in the direction of heat flow, the net heat flow could be too minute to be observed in the DSC curves. It is most probable that the above sample which showed the apparent transition in Fig. 8 has a relatively larger degree of structural relaxation in the amorphous structure compared to the other samples. Both the mentioned structural variations actually occur around −40 °C, although they are hardly detectable in the DSC curves.

The MSEP achieved here is slightly inferior compared to that in our previous publication (0.00059 g cm^{-3}), in which the same samples were measured at 70 °C. The DSC peaks around 70 °C which indicated the pre-melting behaviors were more apparent compared to those of the glass transition found in this study. The larger degree of structural relaxation by pre-melting enhanced the spectral discriminability among the samples further and better accuracy was thereby a consequence. Nonetheless, the enhanced spectral discriminability at

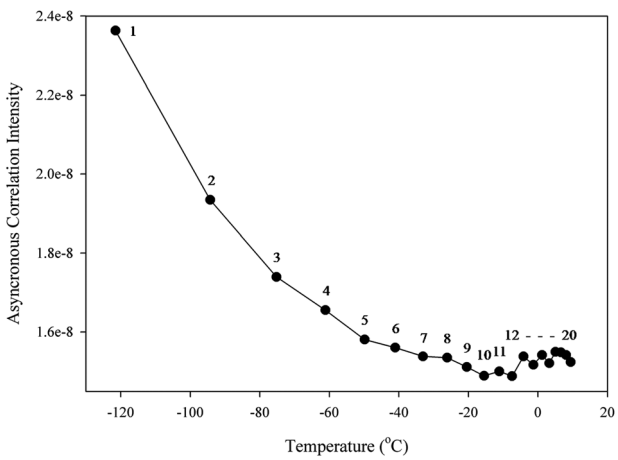

Fig. 7 The variation in the asynchronous correlation intensity between the crystalline and amorphous components as a function of the measurement temperature. The numbers above each circle indicate the corresponding snapshot number.

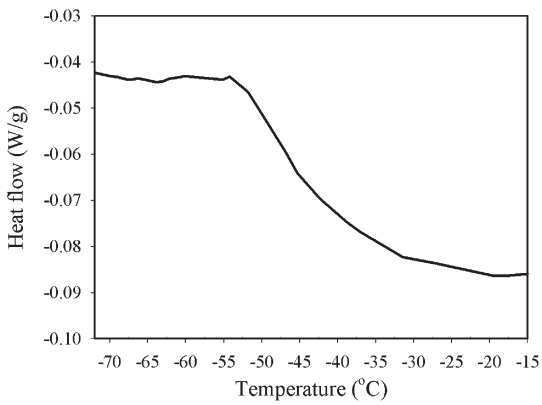

Fig. 8 DSC curve of the low density sample (0.920 g cm^{-3}) in the temperature range of −72 to −15 °C.

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the glass transition temperature also helps to improve the accuracy of the measurement compared to that of the room temperature measurement.

Conclusion

The acquisition of Raman spectra of PE pellets around their T_g enhanced the spectral selectivity among the samples, due to micro-Brownian motions occurring in the amorphous component and minor β-relaxation, and eventually improved multivariate correlation of the spectra with the corresponding sample property (density). 2D correlation analysis effectively described the spectral variations occurring in the amorphous and crystalline components, and provided further insight into the glass transition of PE which has not yet been fully understood. The analytical strategy described here is simple and could be expanded to measure the properties of other polymeric samples, since many polymers have phase transition states such as glass transitions, which could enhance spectral selectivity among samples.

For real application of the strategy, several issues need to be further considered. The inclusion of more samples with diverse grades in a calibration dataset will be necessary for robust routine analysis. In addition, when samples with largely different shapes are measured, possible shape-dependent spectral variation that would degrade the accuracy of the measurements needs to be examined. Since the WAI scheme is used for the acquisition of spectra and the Raman peaks are normalized, no significant shape-dependent spectra are expected when the appearance of the samples are moderately dissimilar, such as for the PE samples in this study. As long as the above-mentioned issues are under control, the proposed method will be versatile for multivariate quantitative spectroscopic analysis of polymers.

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